

TTA-based Expert Routing for Imbalanced Network IDS

Diagnostic Study: Why Stability-based Routing Fails

Lab Meeting

Dataset: CIC-IDS2017 (2.8M samples, 15 classes)

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① Extreme class imbalance

Tail attack classes: Heartbleed, Infiltration, SQL Injection

② Domain shift

Different classes are collected by days

③ OOD overconfidence

⇒ MoE + TTA (Test-Time Aggregation) and OOD detection to mitigate

Limitation of Adopting MATI

- **SADE**: introduced the test-time aggregation (TTA) framework
- **MATI**: extends SADE; one benchmark dataset is *real estate price prediction* ($y \in \mathbb{R}$, continuous)

Our y : class labels $\{0, 1, \dots, 14\}$ — **nominal categorical**

→ No meaningful L2 distance between class labels

Alternative: use prediction stability under perturbation as a routing signal

$$s_e(x) = \frac{1}{1 + \text{JS}(P_e(x) \parallel P_e(x'))}, \quad w_e = \frac{s_e}{\sum_{e'} s_{e'}}$$

Result

TTA weights collapse to near-uniform ($w_e \approx 0.15\text{--}0.18 \approx 1/E$): stability is not a discriminative routing signal. **A different objective loss is needed.**

TTA Formulation — Perturbation

Two perturbation methods to generate x' from x :

(1) Gaussian noise

$$x'_d = x_d + \varepsilon_d, \quad \varepsilon_d \sim \mathcal{N}(0, \sigma \cdot \hat{\sigma}_d)$$

$\hat{\sigma}_d$: per-feature std from training data; σ (noise_std): strength parameter

(2) Feature masking (BERT-style)

$$x'_d = \begin{cases} \hat{\mu}_d & \text{with probability } p_{\text{mask}} \\ x_d & \text{otherwise} \end{cases}$$

Replace each feature independently with its column mean at random

Currently examining numerical behavior across a range of σ and p_{mask} values to validate that these perturbations produce meaningful variation in expert predictions.